

Is Cypress substituting Selenium?

Cypress is very different from Selenium, and comparing the two will be like comparing a bullet train with the metro. Selenium was developed to serve the needs of QA engineers who didn't have access to the code base and chose to write E2E automation tests. Developers like to write tests that are fast, and one pain point for them is the fragility and slowness of creating E2E automation tests. Cypress was created to serve the needs of front-end developers to encourage them to write tests separately from unit tests. Front-end developers mostly use JavaScript. This is why Cypress is built on the JavaScript testing tool. QA engineers, however, can work with different programming languages apart from JavaScript, such as Java or Python.

Some may say that Cypress is just an additional testing tool with features similar to Selenium. But it's well-known that testing is difficult, and creating E2E tests is even tougher. Some developers find that Selenium is not the right tool for them as it's not simple to use it to test asynchronous applications.

However, some QA engineers who are working on a project in JavaScript are still unwilling to switch to Cypress as it does not have some features that Selenium provides.

In my opinion, Selenium is still the boss of automation testing. But with the growing popularity of JavaScript and the focus of test automation management now shifting to developers with the help of QA engineers, Cypress is hurrying to the top.

Installing Cypress

Installing Cypress is an easy task. But we need some prerequisites to do so. We need to have Node.js preinstalled in the system as Cypress is a node based application. Node.js is a JavaScript runtime environment. One can download and install it for a particular operating system from its official resource. The Node.js

installation also tackles the installation of npm (node package manager).

Check the Node.js installation by running the command `node -version` in a terminal. To know the npm version, run the same command.

We also need to have a code editor like Microsoft's Visual Studio Code for suitable programming and preserving project directories.

Installing Cypress with npm

To install Cypress using npm, navigate to the project directory and execute the following command:

```
npm init
```

The command above generates the `package.json` file. Now provide necessary details like `-`, keywords, description, author name and package name.

Once done, run the following command to install Cypress:

```
npm install cypress --save-dev
```

The above command will install Cypress locally as a dev dependency for the particular project.

Wait for some time (it needs to download the binary) and then run the following command:

```
node_modules/.bin/cypress open
```

Cypress will open for the first time. Now you can start writing your test case.

Note: Cypress derives its own folder structure (Figure 1). The folder structure will automatically get generated when Cypress is opened for the first time at that location. It comes with ready-made recipes that show how to test common scenarios in Cypress.

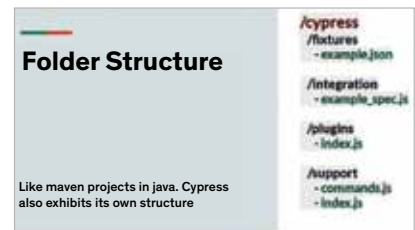


Figure 1: Folder structure

Developers and QA engineers generally keep the test data in a JSON format inside a fixture folder. They write tests inside the integration folder following the same naming convention. Any custom command will come up under the support folder.

To conclude, here are some things to consider:

1. Cypress is a JavaScript tool; this means that even as Selenium works in multiple languages such as Java, Python, C#, etc, Cypress runs only on JavaScript.
2. The tool is mainly meant for the Chrome browser, but now it has cross-browser testing features too.
3. Cypress is not an automation tool; it's a testing tool because it has a test runner Mocha with mocking capabilities and lots of validation rules.
4. It's very fast, and this was done to make the test code itself run in the browser, just as application code does.

This was a basic introduction to the Cypress testing tool. It can be used as a primary E2E tool because it makes life much easier for both developers and testers. Do give it a try if you are interested and working in these fields. Shared below are the links for learning more about it. **END** 🐧

References

- [1] Cypress.io documentation
- [2] <https://github.com/cypress-io/cypress>

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